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#Assessment 2
Part 1 - Data Analysis and Insight
Dataset - Employee Performance

Course: MIS 311 - Introduction to Business Analytics

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1. Data Overview

The dataset selected for this analysis is the Employee Performance dataset, provided as part of the MIS 311 course materials. It represents a typical Human Resources (HR) analytics scenario, capturing demographic, compensation, tenure, and performance information for employees across three departments (HR, IT, and Sales) and three office locations (New York, Los Angeles, and Chicago).

The dataset contains 203 rows and 12 columns before cleaning. Each row represents one employee record. The variables are described below.

According to the dataset description, the dataset consists of the following variables:

Variable	Description
ID	Unique identifier
Name	Names of individuals or products
Age	Age of employees
Department	Name of department
Salary	Monthly salary
Performance Score	Performance evaluation of employee
Experience	Years of experience
Status	If the employee is still working at the company
Location	States
Session	Working session

This dataset is well-suited for business analytics because it combines numerical variables (salary, experience, age, performance score) with categorical variables (department, location, session, status, gender), enabling both descriptive statistics and cross-group comparisons, the kind of analysis an HR or People Analytics team would use to understand compensation fairness, workforce composition, and performance patterns.

2. Data Cleaning

2.1. Missing Value

The Performance Score column had 105 missing values out of 199 cleaned records, equivalent to approximately 52.8%. The remaining columns had no missing values. Due to the high rate of missing values, automatically replacing them with Mean/Median values could distort the actual distribution. Therefore, analyses related to Performance Score were only performed on a sample set with actual data and 94 valid records. It is recommended that businesses standardize their periodic employee performance review process.

2.2. Duplicate Rows

Checked the raw dataset and identified 4 exact duplicate rows. To address this, removed these 4 duplicate records to prevent skewed statistical results, resulting in a clean dataset of 199 unique records.

3. Descriptive Statistics & Visualizations

Table 1 Descriptive Statistics

Age		*Salary*		*Performance Score*		*Experience*	
Mean	37.66316	Mean	5362.116	Mean	2.978947	Mean	10.05263
Standard Error	1.403452	Standard Error	237.7209	Standard Error	0.142745	Standard Error	0.584489
Median	36	Median	4899	Median	3	Median	9
Mode	31	Mode	4899	Mode	2	Mode	7
Standard Deviation	13.67916	Standard Deviation	2317.017	Standard Deviation	1.391302	Standard Deviation	5.696898
Sample Variance	187.1194	Sample Variance	5368568	Sample Variance	1.935722	Sample Variance	32.45465
Kurtosis	-0.76542	Kurtosis	-0.92858	Kurtosis	-1.31171	Kurtosis	-1.19965
Skewness	0.457671	Skewness	0.46861	Skewness	0.038305	Skewness	0.134907
Range	47	Range	7916	Range	4	Range	19
Minimum	18	Minimum	2025	Minimum	1	Minimum	1
Maximum	65	Maximum	9941	Maximum	5	Maximum	20
Sum	3578	Sum	509401	Sum	283	Sum	955
Count	95	Count	95	Count	95	Count	95

The average employee is approximately 38 years old with 10 years of work experience. The average monthly salary is USD 5,362, while the average performance score is 2.98 out of 5.

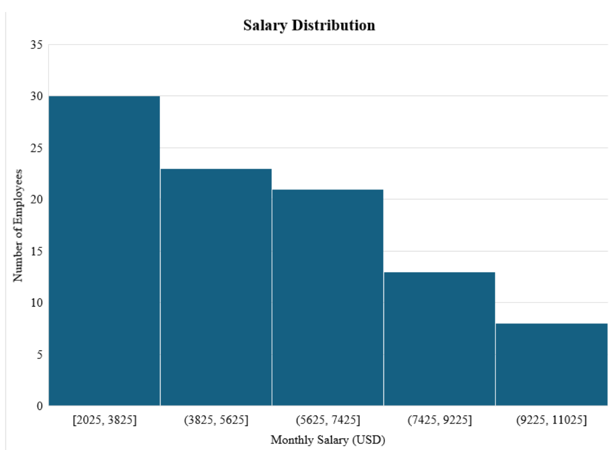

Figure 1 Salary Distribution

The histogram shows that employee salaries are concentrated in the lower and middle salary ranges. The number of employees gradually decreases as salary increases, indicating that relatively few employees receive very high salaries.

Row Labels	**Average of Performance Score**
HR	2.852941176
IT	3.285714286
Sales	2.848484848
Grand Total	**2.978947368**

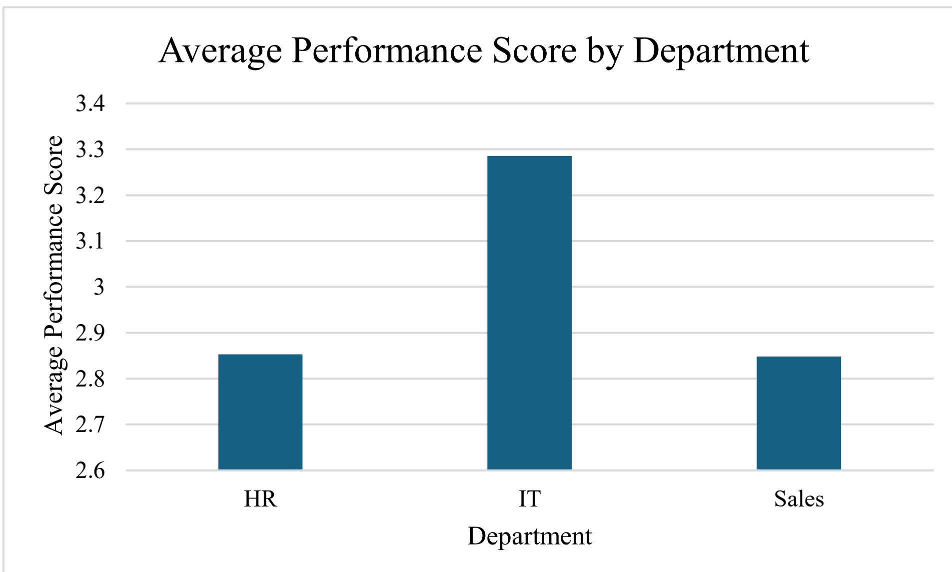

Figure 2. Average Performance Score by Department

Employees in the IT department have the highest average performance score, while HR and Sales have similar average scores. However, the differences among departments are relatively small, indicating that employee performance is generally consistent across the organization.

Row Labels	**Average of Salary**
HR	5219.647059
IT	5624
Sales	5286.69697
Grand Total	**5362.115789**

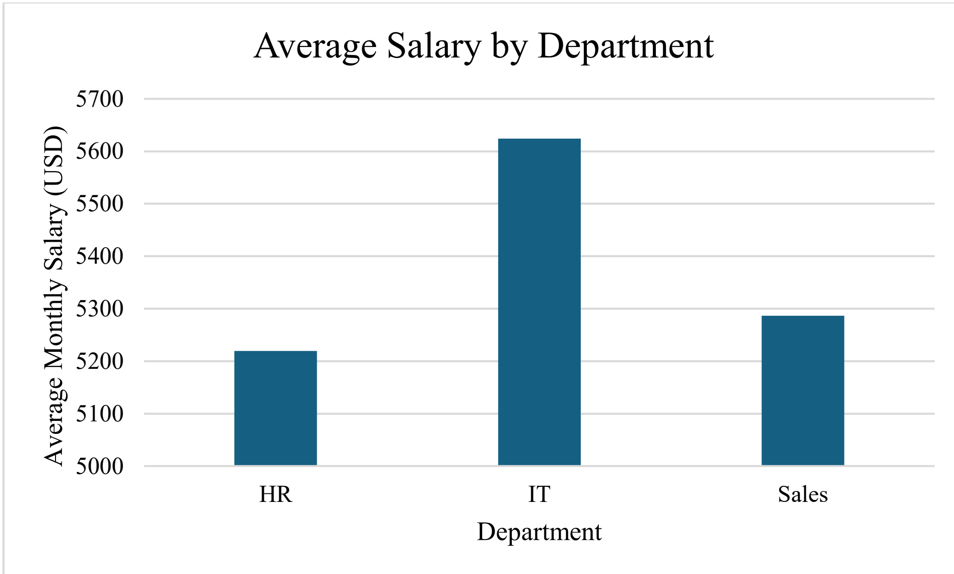

Figure 3. Average Salary by Department

The IT department has the highest average salary, while HR has the lowest average salary. This suggests that employees in technical roles are compensated more highly, likely because they require specialized knowledge and skills.

Insight 1: The IT department demonstrates the strongest performance and receives the highest compensation.

The IT department has the highest average salary $5,624 and the highest average performance score 3.29, both exceeding the company averages of $5,362.12 and 2.98, respectively. This suggests that the company's investment in IT talent is associated with higher employee performance, highlighting the strategic importance of technical expertise to the organization.

Insight 2: Higher salaries do not always translate into better performance across departments.

Although the Sales department earns a slightly higher average salary $5,286.70) than the HR department $5,219.65, both departments achieve almost identical performance scores of 2.85. This indicates that salary alone is not the primary driver of employee performance. Other factors, such as training, work environment, job complexity, and performance management practices, may have a greater influence on employee productivity.